DATHEON INTERNSHIP PROJECT

TASK 1
Exploratory Data Analysis (EDA) on Superstore Retail Sales

A practical study of retail sales, profitability, product mix and regional performance

SUBMITTED BY	INSTITUTION	DOMAIN
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Programme: Datheon Data Analytics & Power BI Internship | Batch 01
Dataset disclosure: synthetic demonstration dataset prepared for reproducible portfolio practice.

1. Executive Summary

This project demonstrates a complete exploratory data analysis workflow on a retail-style dataset containing 1,202 raw records. The workflow profiles data quality, removes duplicate records, handles missing categorical and postal-code values, engineers date and margin fields, and evaluates sales and profit by time, region and category. After removing duplicate records, the analytical dataset contains 1,200 rows.

Metric	Result
Total Sales	$1,015,888.41
Total Profit	$183,203.25
Distinct Orders	1,200
Average Order Value	$846.57
Overall Profit Margin	18.03%
Rows after cleaning	1,200

2. Objectives

• Understand dataset structure and business variables.
• Detect missing values, duplicates and data-type issues.
• Clean the dataset using transparent, reproducible rules.
• Summarize sales, profit, discounts and order patterns.
• Communicate findings through charts and actionable recommendations.

3. Dataset and Methodology

The dataset includes order and shipping dates, customer and segment attributes, geography, product hierarchy, sales, quantity, discount and profit. It is a synthetic Superstore-style dataset generated for this project. It should not be described as Datheon's official starter file. The workflow uses Python, pandas, NumPy and Matplotlib.

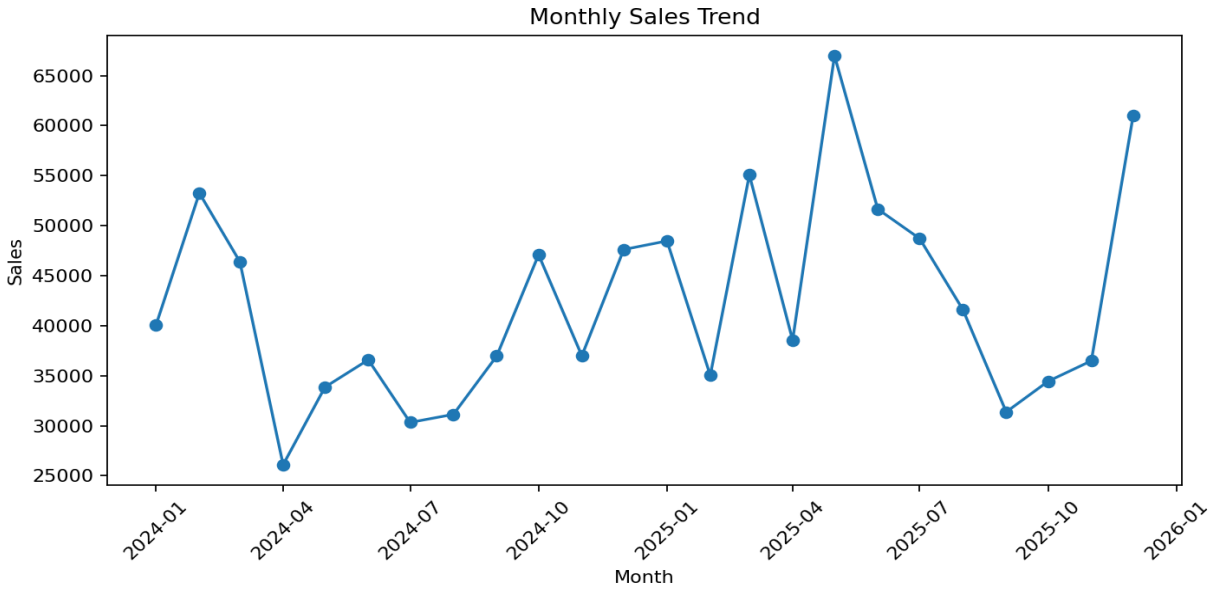
Stage	Method
Load & profile	Read CSV; inspect dimensions, columns, data types and null counts.
Quality checks	Count duplicate rows; review missing values and numeric distributions.
Cleaning	Remove exact duplicate rows; fill missing postal codes with 0, ship mode with mode, and customer name with Unknown
Feature engineering	Create Order Year, Order Month, Shipping Days and Profit Margin.
Analysis	Aggregate sales/profit by month, region, category and product; examine discount-profit relationship.
Export	Save cleaned data and chart-ready summary CSVs.

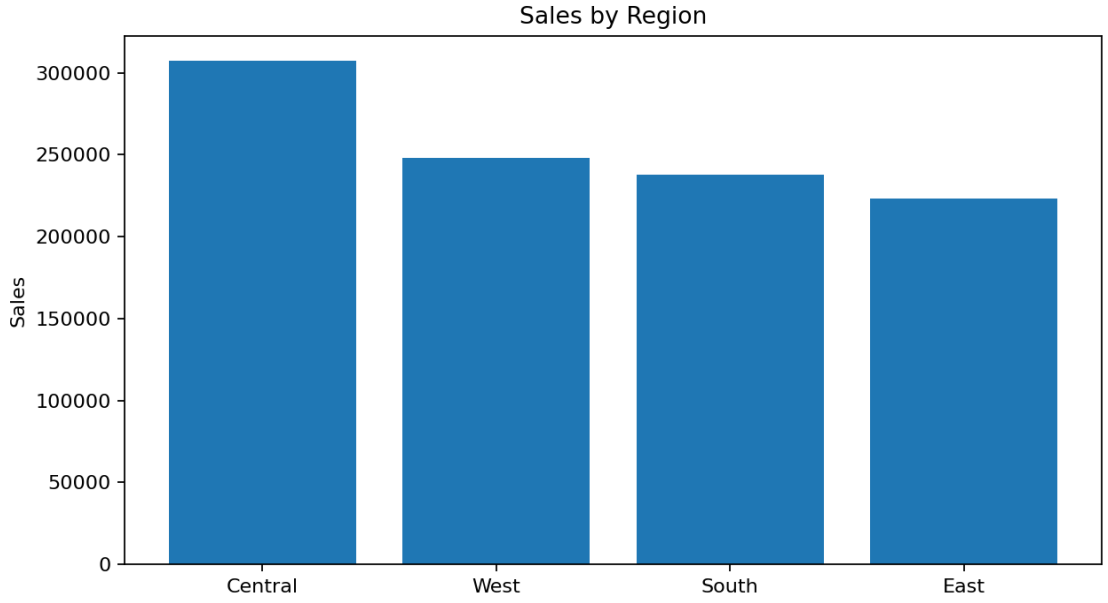
4. Data Quality and Cleaning

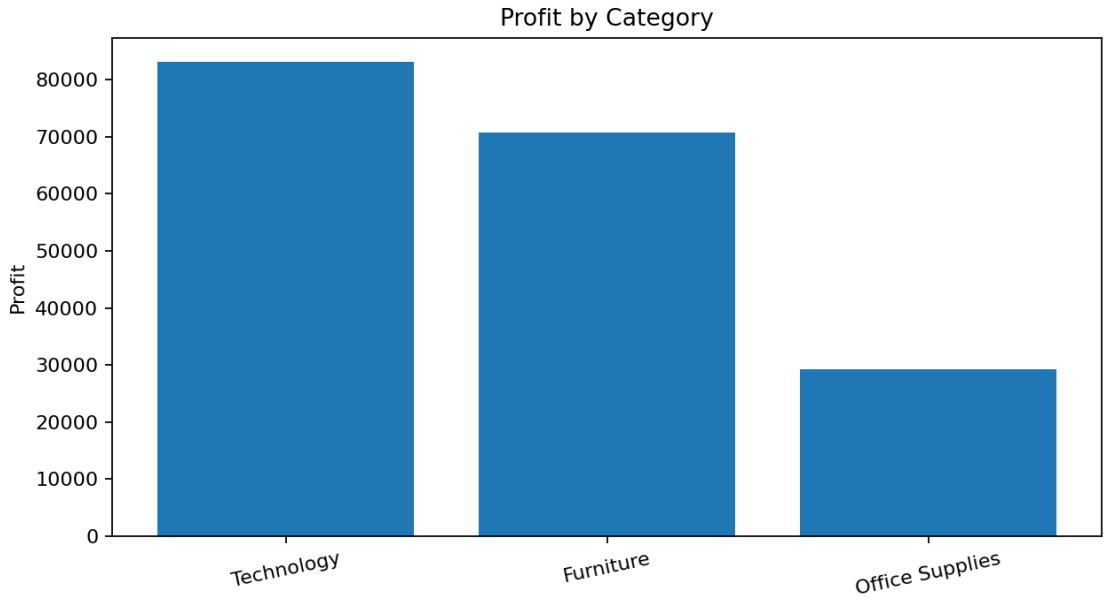
Raw rows: 1,202. Duplicate rows detected and removed: 2. Missing values were intentionally introduced in a small number of fields to demonstrate data-quality handling. Postal Code nulls were set to 0 (unknown), missing Ship Mode values were replaced with the most frequent mode, and missing Customer Name values were labelled Unknown Customer. These choices preserve rows while making assumptions explicit.

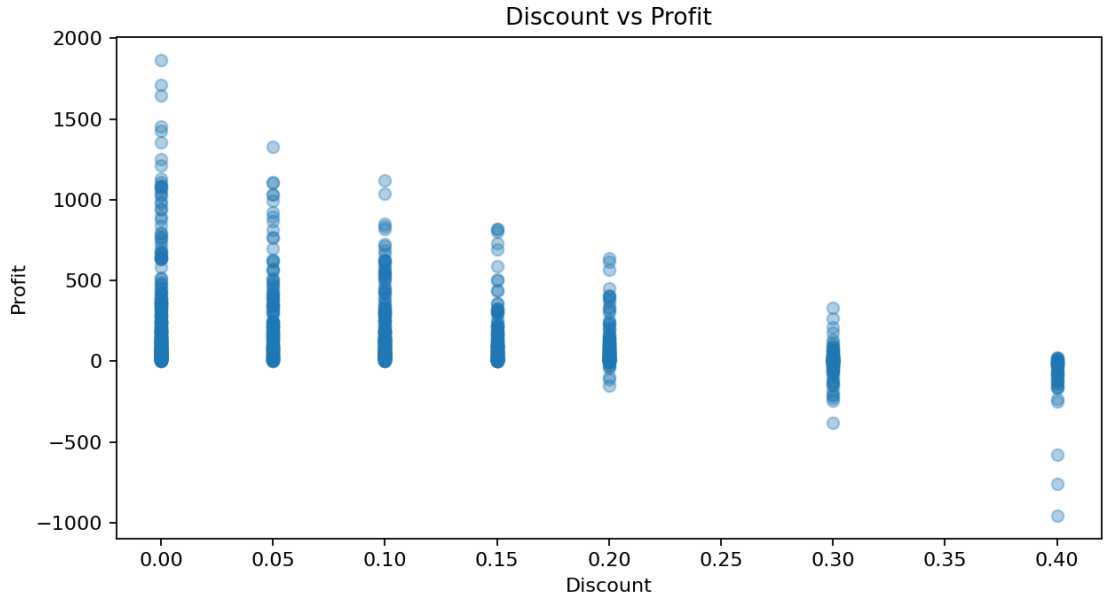
5. Exploratory Findings

The strongest sales region in this generated sample is Central ($307,187.87). The leading sales category is Technology ($518,187.20). Profit is highest in Technology ($83,178.39). These findings describe only this synthetic dataset and should be recalculated if an official dataset is supplied.


Monthly Sales Trend
Sales
65000
60000
55000
50000
45000
40000
35000
30000
25000
2024-01
2024-04
2024-07
2024-10
2025-01
2025-04
2025-07
2025-10
2026-01
Month


Sales by Region
Sales
300000
250000
200000
150000
100000
50000
0
Central
West
South
East





6. Business Recommendations

1. Review regional performance monthly and investigate regions with high sales but comparatively low profit.
2. Evaluate product-level contribution rather than relying on revenue alone.
3. Monitor discount bands alongside profit to identify discount levels that erode margin.
4. Use the monthly trend to plan inventory and promotional calendars.
5. Track shipping duration and ship mode to assess service efficiency.

7. Limitations

This is a synthetic practice dataset and does not establish real-world customer behaviour, company performance or causal relationships. Postal code 0 is a placeholder for missing values. Correlation and visual patterns should not be interpreted as proof of causation. Replace the synthetic file with Datheon's required starter dataset if one is provided.

8. Deliverables and Reproducibility

Source code: task1_eda/src/run_eda.py
Raw dataset: datasets/superstore_retail_raw.csv

Clean dataset: datasets/superstore_retail_clean.csv
Charts and summary tables: task1_eda/outputs/
Run command: python task1_eda/src/run_eda.py

9. Conclusion

The project demonstrates the essential EDA lifecycle: data profiling, cleaning, feature creation, aggregation, visualization and business interpretation. The outputs provide a reproducible foundation for the executive dashboard in Task 2.